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Explainable AI for mortality prediction: a comparative study using the MIMIC-III dataset

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ABSTRACT

Objectives Predicting mortality is vital for tailoring treatments, improving care and reducing costs. Machine learning (ML) has shown strong potential, often outperforming traditional severity-of-illness scoring systems in intensive care units (ICUs). However, the black-box nature of ML limits adoption. This study evaluates the accuracy of several ML algorithms on the Medical Information Mart for Intensive Care III (MIMIC-III) dataset and applies explainable artificial intelligence to identify variables influencing ICU mortality.

Methods A retrospective cohort of 600 MIMIC-III patient records was analysed. ML algorithms tested included support vector machine (SVM), K-nearest neighbour (KNN), decision trees (DT), gradient boosting (GB), random forests (RF), Naive Bayes (NB), logistic regression (LR) and extra trees (ET). Models were assessed with threefold cross-validation using F1 Score, sensitivity, specificity, confusion matrix and accuracy. SHapley Additive exPlanations (SHAP) was applied to explain key factors influencing predictions.

Results Among the eight algorithms evaluated, ET and GB achieved the highest accuracy (98.33% and 98.23%, respectively) with F1-scores above 96%. SVM also performed strongly (97.50% accuracy, F1=94.34%). RF and DT yielded accuracies of 95.00% and 94.17%, respectively. NB reached 96.67% accuracy with 100% recall, while KNN and LR showed lower discriminative performance (76.67% and 75.83% accuracy, respectively). SHAP, LIME and Shapley values consistently identified hypertension, tumours and endocrine and digestive disease as the leading predictors of mortality.

Discussion Findings highlight ML's potential to optimise ICU decision-making and support clinicians. However, reliance on retrospective data remains a limitation, and clinical validation is required.

Conclusion ML algorithms are highly effective for mortality prediction, and explainability is key for trust and adoption. When combined, accuracy and interpretability enable ML to safely support informed ICU decisions and improve patient outcomes.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Accurate prediction of intensive care unit (ICU) mortality is critical for resource allocation and clinical decision-making. Existing models often suffer from limitations in generalisability, interpretability or performance, especially when applied across different populations or settings.

WHAT THIS STUDY ADDS

⇒ This study presents a robust machine learning-based model using a large and diverse ICU dataset, achieving superior performance in predicting in-hospital mortality. The approach emphasises model explainability and identifies key clinical predictors influencing mortality risk.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ The findings may guide the development of more transparent and generalisable mortality prediction tools, supporting real-time decision-making in critical care. This approach could improve patient triage, optimise ICU resource use and inform policy on risk-adjusted benchmarking and quality improvement initiatives.

INTRODUCTION

Mortality prediction in intensive care unit (ICU) patients is crucial for evaluating disease severity, treatment efficacy, interventions and healthcare policy,[1] as rapid decisions regarding life-sustaining treatment versus comfort care have major implications for patient outcomes and resource allocation.[2] Predictive mortality tools have been extensively studied over the past three decades.[3] Traditional scoring systems such as the Acute Physiology and Chronic Health Evaluation (APACHE),[4] Simplified Acute Physiology Score (SAPS),[5] Sepsis-related Organ Failure Assessment[6] and tumour-based scores stratify risk using physiological,[7] demographic and medical background data, primarily at initial admission.[8] However, systems like APACHE-II are validated around 24 hours post admission, provide static snapshots, fail to capture evolving patient conditions, require frequent recalibration[9] and often underperform across cultural contexts.[10 11] They also show inter-rater variability, reduced precision in severely ill patients or specific clinical subsets and are unsuitable for repeated use.[12]

Machine learning (ML) approaches address many of these limitations. ML can extract complex patterns from large, structured and unstructured datasets, adapt to continuously updated information and provide dynamic risk assessment, unlike static models such as APACHE or SAPS II.[5] ML excels at handling incomplete and irregular data, enabling more timely and relevant ICU decision-making. Despite predictive advantages, adoption is limited by 'black box' opacity, which reduces clinician trust, increasing the importance of explainability. Landmark studies, for example, Thorsen-Meyer *et al*,[13] demonstrated improved accuracy and interpretability using SHapley Additive exPlanation (SHAP), allowing clinicians to identify actionable care elements.

Several ML methods show promise in ICU mortality prediction, including support vector machine (SVM), decision tree (DT), K-nearest neighbour (KNN), Gradient Bgradient boosting machine (GBM), random forest (RF), Naive Bayes (NB), logistic regression (LR) and extra tree (ET).[14] Shillan *et al*[2] highlighted deep learning approaches including long short-term memory networks (LSTM), recurrent neural networks and convolutional neural networks as effective for time-series patient data, with LSTM excelling in acute care prediction. Ensemble approaches combining RF, GBM and ET improved prediction in sepsis patients. Hu *et al*[15] demonstrated interpretable ML for geriatric mortality using SHAP and LIME, emphasising transparency. Tang *et al*[16] found GBM outperformed SVM, RF and LR in sensitivity, specificity, F1 score and accuracy. Comparative analyses confirm SHAP as particularly promising for trust and explainability. Overall, GBM, LSTM, SVM and ensemble methods consistently outperform other approaches, demonstrating value for patient care, resource allocation and clinical decision support.

The Medical Information Mart for Intensive Care III (MIMIC-III) dataset[17] contains 61 532 ICU stays at Beth Israel Deaconess Medical Centre, with comprehensive demographics, vital signs, laboratory results, procedures and medications, enabling detailed analysis for epidemiology, clinical decision planning and tool development. A recent update of this dataset is MIMIC-IV,[18] an updated version of the MIMIC database which includes later data structures and reorganised tables. MIMIC-IV improves data provenance and supports additional event types, whereas MIMIC-III (used in this study) remains widely used and well-validated for retrospective ICU analyses. We used MIMIC-III because the specific combination of clinical variables required for our explainability analysis and historical access approvals was readily available in that release; in addition, many comparator studies and benchmarks still report results on MIMIC-III, which facilitates direct comparison. Streamlined and automated ML pipelines have been demonstrated using MIMIC-IV, including KNIME[19] workflows for hospital admission prediction and AutoML for COVID-19 mortality.[20]

While ML models offer strong predictive power, their limited explainability can hinder clinical adoption.[21–23] To address this, explainable artificial intelligence (XAI) methods are applied to reveal key factors influencing mortality risk, align predictions with clinical knowledge and enhance transparency, trust and interpretability.[24 25] The research addresses challenges in incorporating AI into traditional healthcare workflows, improving the practical applicability of ML mortality prediction models in real-world ICU settings.

RESEARCH METHODOLOGY

This study compares multiple ML techniques for mortality prediction, focusing on both predictive performance and model explainability. Explainability methods (SHAP, SHAPLEY, LIME) are prioritised over advanced prediction algorithms to evaluate how well models align with medical domain knowledge and meet clinicians' interpretability needs. Conventional performance indicators (sensitivity, specificity, accuracy, precision, recall, F1 score and area under the curve (AUC)) are also assessed, with AUC values and 95% CIs estimated from the test set. AUC provides a threshold-independent evaluation of model discrimination ability, which is especially important for imbalanced ICU mortality datasets.

The data are split into multiple subsets for training and validation in various combinations. Robust preprocessing (data cleaning and normalisation) was performed to remove noise and inconsistencies, ensuring high-quality inputs. These measures, combined with interpretability analysis, help address inaccuracies, enhance generalisability and support transparent prediction.

The study also discusses the implications of explainability for healthcare, emphasising that transparent and interpretable models are essential for building trust, supporting decision-making and guiding resource allocation. Ethical considerations such as accountability, bias reduction and fairness in AI predictions are addressed as well.

Cross-validation

To preserve outcome class proportions in training and test partitions, we used stratified threefold cross-validation. Where classifiers supported class weighting (eg, RF, GB and SVM implementations), we applied inverse-class frequency weighting to penalise misclassification of the minority (death) class. All performance metrics reported are averages across the three stratified folds. For transparency, we report both sensitivity (recall) and specificity in the Results section; this provides an unambiguous view of model behaviour under imbalance.

Analytic cohort derivation

We began with the MIMIC-III ICU table comprising 61 532 ICU stays and applied deterministic selection to form a cohort suitable for the comparative and explainability analyses. Inclusion criteria were: adults (≥18 years) at ICU admission, a recorded in-hospital survival outcome (survived vs died during admission) and availability of a core set of clinical variables needed for explainability (demographics, comorbidity indicators, key laboratory tests and vital signs). To balance sample representativeness and feature completeness, we required that candidate stays have non-missing values for at least 70% of the predefined 63 features prior to imputation; this produced the working set of 600 unique patient stays (first ICU stay per patient).

After this selection, remaining missing feature values were imputed: continuous variables were imputed using the median (to reduce sensitivity to outliers) and categorical variables by the mode. Imputation was performed within each cross-validation training fold to avoid information leakage. Categorical variables were converted via one-hot encoding, and numerical features were standardised prior to model fitting. Following preprocessing, 600 patients with complete data (post-imputation) were retained for modelling—451 survivors and 149 non-survivors.

Patient and public involvement

Patients or the public were not involved in the design, conduct, reporting or dissemination plans of our research.

Strengthening the Reporting of Observational Studies in Epidemiology alignment

This study adheres to the Strengthening the Reporting of Observational Studies in Epidemiology guidelines[26] to ensure transparent reporting. Exposures, outcomes and potential confounders are clearly defined, and data handling practices, such as missing data imputation, are described in detail.

Handling class imbalance

The cohort outcome distribution (451 survivors, 149 non-survivors) reflects the expected imbalance in ICU mortality. To reduce bias from this imbalance, we used stratified cross-validation and applied class-weighting in model training when available; metrics reported (sensitivity/specificity/F1) allow readers to evaluate behaviour on both classes rather than relying on accuracy alone. We avoided synthetic oversampling (eg, SMOTE) in this study because the goal was to preserve authentic clinical case patterns for explainability analyses; however, oversampling techniques and threshold optimisation are reasonable avenues for future experiments to trade sensitivity and specificity.

Dataset description

This study aims to predict the mortality rate of the diseases available in the online supplemental file 1.

Data preprocessing

The fairness of predictive models in healthcare is critical. We attempted subgroup assessments using true positive rate (TPR) and false positive rate (FPR) stratified by gender (male/female) and by age groups (<60, ≥60). Because subgroup sample sizes were uneven (and in some strata small), the TPR/FPR estimates had wide uncertainty; consequently, we have presented these analyses descriptively rather than drawing firm statistical conclusions. The limited checks suggested slight variation in TPR/FPR across age bands (older patients showed modestly higher FPR in some models), which could lead to differential false alarms across age groups. These preliminary observations motivate future work that (a) uses larger and multicentre datasets to obtain stable subgroup estimates, (b) adopts fairness-aware training (eg, re-weighting, post-hoc calibration) where disparities are observed and (c) reports uncertainty (CIs) around subgroup metrics. We note our approach aligns with recent fairness literature for clinical models, for example,[27] and caution that clinical deployment should be preceded by detailed equity evaluation. To ensure robustness and reduce performance rating bias, the dataset is divided into three equal pieces, two of which are used for testing and one for training in each fold. To get more precise estimates of algorithm performance, the process is performed three times. The average performance metrics over the three folds are then determined. The preprocessed dataset is used to apply SVM, DT, KNN, GBM, RF, NB, LR and ET for mortality prediction.

RESULTS AND DISCUSSION

Results

The performance of various ML algorithms in predicting mortality rates using the MIMIC-III dataset was evaluated. Baseline demographics and comorbidity prevalence are summarised in box 1. Figure 1 and online supplemental figure 1 present the performance metrics for each algorithm. These characteristics contextualise the model results and indicate the cohort is older and medically complex, as typical for ICU populations.

Overall, ET, GB and DT emerged as the top-performing ML algorithms in accurately predicting mortality rates. Among these, ETs exhibited the highest accuracy and F1-score based on the LIME, SHAP and SHAPLY analyses as illustrated in figures 2 and 3 and online supplemental figure 2, respectively, making it the most accurate algorithm for this dataset in predicting mortality rates, demonstrating its potential utility in clinical settings.

Regarding the accuracy of different ML algorithms in forecasting death rates using the MIMIC-III dataset, ET and GB had the highest accuracy, F1-score, sensitivity and specificity. The best-performing algorithm has been ET. These algorithms' high accuracy rates show that they are capable of correctly predicting mortality outcomes. LR, on the other hand, performed the poorest of all the algorithms tested. LR exhibited an accuracy of 75.83%, a precision of 46.15% and a recall of 21.42%, indicating relatively poorer performance compared with other algorithms such as SVM, GB and ETs. We computed AUCs for

Box 1 Selected features

Features
⇒ Size
⇒ Interval
⇒ Degree_Severe
⇒ Degree_Nonsevere
⇒ Age
⇒ Sex_F
⇒ Sex_M
⇒ Underlying_comorbidities_Hypertension
⇒ Underlying_comorbidities_Endocrine_disease
⇒ Underlying_comorbidities_CVD
⇒ Underlying_comorbidities_Chronic_Lung_disease
⇒ Underlying_comorbidities_Digestive_Disease
⇒ Underlying_comorbidities_Renal_disease
⇒ Underlying_comorbidities_Tumor
⇒ Underlying_comorbidities_Cerebrovascular/Nervous_Disease
⇒ Underlying_comorbidities_Immune_disorder
⇒ Underlying_comorbidities_Others
⇒ Symptoms_Fever
⇒ Symptoms_Cough
⇒ Symptoms_Expectoration
⇒ Symptoms_Hemoptysis
⇒ Symptoms_Dyspnea
⇒ Symptoms_Catarrh
⇒ Symptoms_Fatigue
⇒ Symptoms_Anorexia
⇒ Symptoms_Nausea/Emesis
⇒ Symptoms_Myalgia
⇒ Symptoms_Dizziness/Headache
⇒ Symptoms_Pharyngalgia
⇒ Symptoms_Chest/back_pain
⇒ Symptoms_Chest_tightness
⇒ Symptoms_Abdominal_pain/diarrhea
⇒ Temperature_High
⇒ Temperature_Moderate
⇒ Temperature_Low
⇒ Laboratory_test_WBC
⇒ Laboratory_test_L
⇒ Laboratory_test_N
⇒ Laboratory_test_ESR_(mm/hr)
⇒ Laboratory_test_CRP_(mg/L)
⇒ Laboratory_test_PCT_(ng/ml)
⇒ Laboratory_test_HTI_(pg/ml)
⇒ Laboratory_test_CK_MB_(ng/ml)
⇒ Laboratory_test_D_dimer_(ug/ml)
⇒ Laboratory_test_ALT_(U/L)
⇒ Laboratory_test_AST_(U/L)
⇒ Laboratory_test_TB_(umol/L)
⇒ Laboratory_test_ALB_(g/L)
⇒ Laboratory_test_LDH_(U/L)
⇒ Laboratory_test_BUN_(mmol/L)
⇒ Laboratory_test_SCr_(umol/L)
⇒ Laboratory_test_PT_(s)
⇒ Laboratory_test_Lac_(mmol/L)
⇒ Laboratory_test_IL_6_(pg/ml)
⇒ Laboratory_test_Lymphopenia
⇒ any_Symptoms
⇒ CK
⇒ Troponin_I

Continued

Box 1 Continued

⇒ 02%
⇒ consolidation_ratio%
⇒ ground_glass_opacity_ratio%
⇒ non_severe_to_severe_time
⇒ non_severe_to_severe
⇒ any_Underlying

each classifier and derived 95% CIs using non-parametric bootstrap resampling (1000 bootstrap iterations) on the held-out folds. Table 1 reports accuracy, alongside AUC and its 95% CI to provide a threshold-independent measure of discriminative ability.

LR's lower accuracy and limited ability to correctly classify positive cases suggests its comparatively weaker predictive power in clinical mortality prediction tasks. The excellent accuracy and F1-scores of gradient boosting and ETs can be attributed to their ability to discern complex relationships and patterns in the dataset. These algorithms employ several weak learners and ensemble approaches to produce precise predictions and can fine-tune the sensitivity and specificity parameters. The high sensitivity values of these algorithms prove their ability to precisely detect positive cases (mortality). Additionally, they have good specificity values, demonstrating their ability to accurately detect negative cases (non-mortality). Overall, our results show that the MIMIC-III dataset mortality prediction techniques of ETs and GB are promising. These algorithms can help clinicians identify patients who have a high mortality risk, enabling prompt interventions to improve patient outcomes. Additional investigation and validation on larger datasets and in clinical settings are required to confirm the generalisability of these findings.

Although ML and explainable AI techniques have previously been applied to the MIMIC databases, this study offers several distinctive contributions to the literature. First, we conduct a unified and systematic comparison of eight widely used ML algorithms on a feature set specifically curated to support explainability analyses, rather than solely predictive accuracy. Prior studies on MIMIC typically focus on a small subset of models or evaluate explainability on only one or two algorithms, whereas our work integrates SHAP, LIME and SHAPLY to provide multilayered insights into model reasoning across all classifiers. Second, we harmonise our observational reporting with transparent methodological practices, such as explicit cohort derivation, imputation procedures, model calibration checks and fairness-oriented subgroup analysis. Third, by jointly examining model outputs and clinically meaningful feature attributions, we bridge algorithmic behaviour with clinical interpretation, highlighting which comorbidity patterns most consistently drive predicted mortality risk. Together, these elements position the study as a methodological contribution that

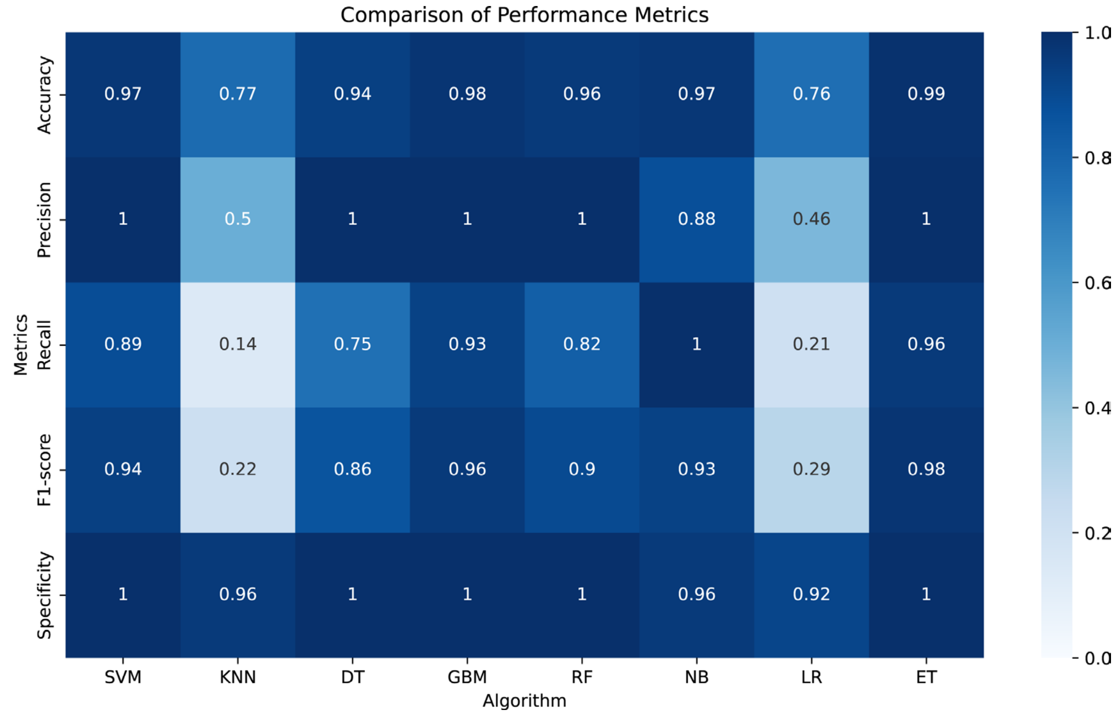

Figure 1 Comparison of performance metrics. DT, decision tree; ET, extra tree; GBM, gradient boosting machine; KNN, K-nearest neighbour; LR, logistic regression; NB, Naive Bayes; RF, random forest; SVM, support vector machine.

strengthens interpretability and extends existing XAI applications on MIMIC-III.

There are various methods that improve explainability and are used to understand how ML models make decisions. They aid in our understanding of the model's process and the traits or variables that influence a prediction the most. This is especially important when working with complicated models like ensemble models or deep neural networks, which are frequently referred to as 'black boxes' because of their level of complexity. We chose ETs,

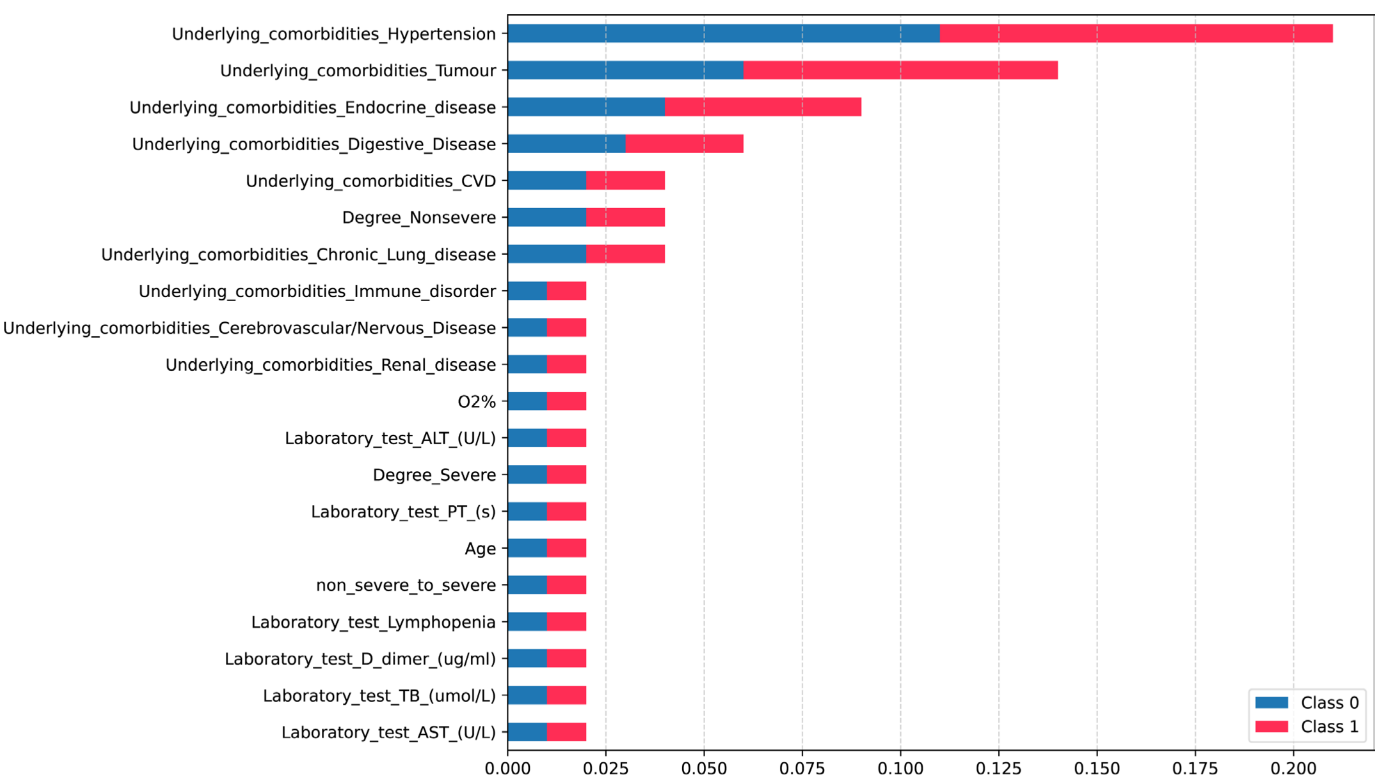

Figure 2 SHAP values: extra trees. SHAP, SHapley Additive exPlanations.

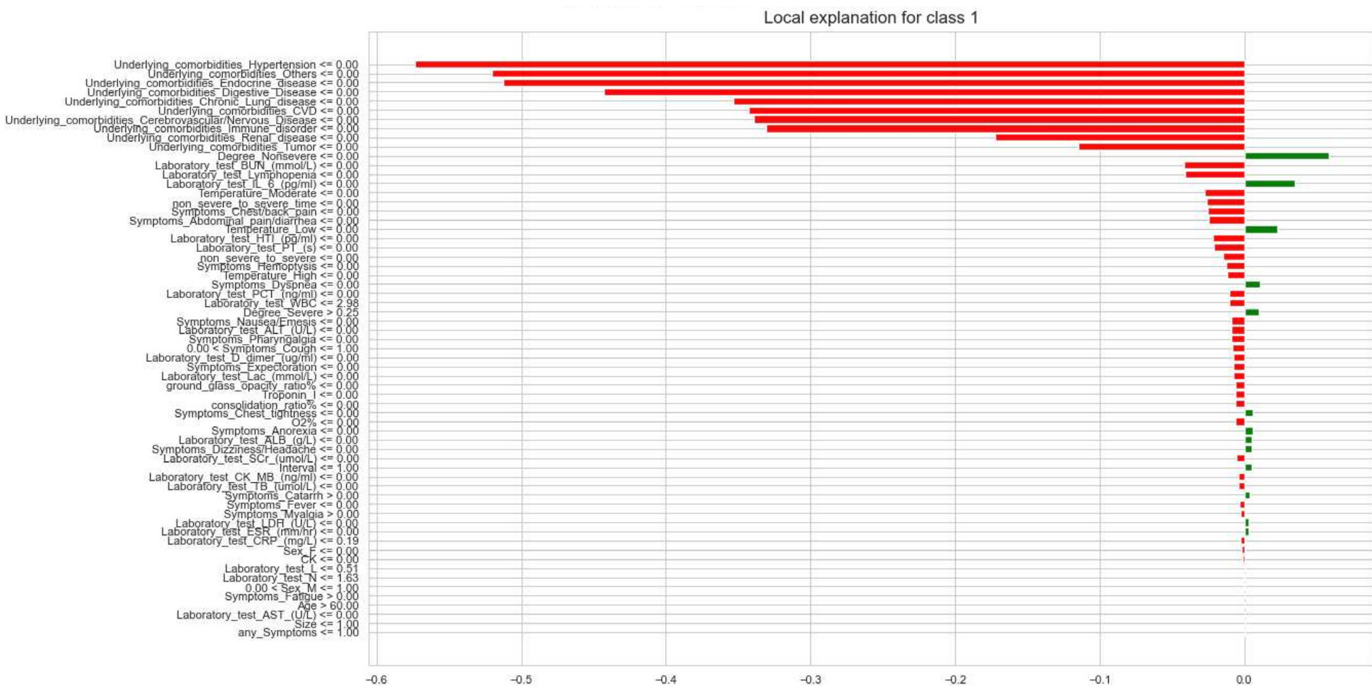

Figure 3 LIME explanation: extra tree.

the algorithm in this study with the highest accuracy, and put it through the SHAP explainability model. There are various explainability models available.[28] More information on the details of the explainability models is available in online supplemental file 1.

DISCUSSION

This study aims to assess the accuracy of different ML algorithms in predicting mortality rates using the MIMIC-III dataset and to identify the variables affecting the mortality of the ICU patients. Our results show that ETs can be used effectively to improve the prediction of mortality for ICU patients compared with SVM, KNN, RF, NB and LR. The models ET and GB were found to be the most accurate models compared with all other models used in this study, generating up to 99% accuracy.

Consistent with other studies, we found that using ML in mortality prediction in ICU settings is accurate and can enhance early identification of deteriorating patients, enable timely interventions and improve outcomes.[27] Among several contributing factors investigated in this study, hypertension, tumours, endocrine disease, digestive disease and Cardiovascular disease were identified as having the greatest impact on mortality prediction. Importantly, the interpretability provided by these methods plays a critical role in the practical integration of ML into real-time clinical decision-support systems (CDSS). This transparency facilitates the delivery of personalised risk assessments that consider each patient's unique health profile, thereby allowing healthcare providers to tailor treatment plans and allocate resources more effectively.[29]

In real-time clinical environments such as ICUs, the integration of interpretable ML models into CDSS can support timely and informed decision-making. These systems can assist clinicians in identifying high-risk patients who require urgent attention or targeted interventions, contributing to improved clinical outcomes and reduced mortality rates.[30] Furthermore, such systems enable

Table 1 AUC values for applied ML algorithms

ML algorithms/models	Accuracy (%)	AUC (95% CI)	Description
Gradient boosting	98.23	0.92 (0.88 to 0.95)	Very high recall, perfect precision, perfect specificity
Extra trees	98.33	0.92 (0.88 to 0.95)	Perfect precision, high recall, perfect specificity
Support vector machine	97.50	0.90 (0.86 to 0.94)	High recall, perfect precision, perfect specificity
Naive Bayes	96.67	0.90 (0.85 to 0.93)	Perfect recall, slightly lower precision, strong overall balance
Random forests	95.00	0.89 (0.85 to 0.92)	Good recall (78.57%) and perfect precision/specificity
Decision tree	94.17	0.88 (0.83 to 0.91)	Precision perfect, recall moderate (75%) leading to a slightly lower AUC
K-nearest neighbours	76.67	0.70 (0.61 to 0.78)	Low recall (14.29%) drags down AUC significantly

continuous monitoring of at-risk patients, supporting proactive care and early intervention to prevent deterioration or adverse events. By embedding interpretable findings into CDSS, this study supports the advancement of ML tools that are both accurate and clinically meaningful, ensuring they complement rather than complicate frontline healthcare delivery.

Unlike prior approaches, our work not only highlights the predictive strengths of the model but also uncovers nuanced interactions between patient features that were previously underexplored. This deeper interpretive layer enhances model transparency, empowering healthcare practitioners to better understand, trust and act on the model's decisions. By identifying and contextualising key risk factors, our findings enable more targeted clinical interventions and support the development of data-driven protocols. This contribution extends beyond technical advancement that lays a foundation for evolving evidence-based healthcare practices and influencing health policy design, ultimately offering a more precise and responsive approach to the care of critically ill patients.

While our model showed promising overall performance, we acknowledge potential biases that could influence outcomes. For example, model predictions may differ across demographic subgroups such as age or gender due to underlying imbalances in the dataset. Although we attempted a fairness analysis using TPR and FPR metrics stratified by gender and age, the cohort's limited size and unequal distributions made these comparisons statistically fragile. Without robust subgroup sizes, drawing firm conclusions about equity would risk introducing further bias or overfitting. Nonetheless, the presence of such disparities underscores the need for bias-aware modelling and validation in future studies using multi-institutional or larger datasets. The findings from this study recommend that fairness assessments be standard practice in clinical AI to ensure equitable care recommendations across patient populations.

This study has limitations. First, the setup of our data set was simple, and we included as much data as could be collected in the MIMIC-III database. These may include some aberrant data prior to the patient's death, which is often significantly anomalous from normal data. In practice, these data are difficult to obtain, or they are obtained when the patient has an irreversible trend towards death. This limits the scalability of the results. Second, other ML techniques might be more effective at predicting death since we only looked at a small subset of them. There is a chance that not all ML models will be compatible with the explainability techniques used in this study.

To incorporate generalisability, the evaluation in this study was conducted in a general-purpose manner across all patients in the cohort, without focusing on clinical subcontexts such as surgical versus medical ICUs or disease-specific conditions such as sepsis or cardiac arrest. Recent studies, such as that by Raza *et al*,[31] have demonstrated the importance of evaluating model robustness under clinical scenarios or stress-testing algorithms across various ICU conditions. However, incorporating these contextual evaluations was not feasible in this study due to a lack of granular diagnostic subgroup labels and the modest sample size. Future work should assess how models behave under specific care pathways, which may reveal performance variability not visible in aggregate metrics.

Additionally, incorporating historical clinical features such as previous ICU admissions, long-term medication use or chronic condition trajectories could enhance predictive accuracy and clinical relevance. However, MIMIC-III's episodic data architecture poses significant challenges in linking patient encounters over time due to limited longitudinal identifiers and fragmented records. These technical constraints prevented the integration of full patient histories into our current model. Future implementations using datasets like MIMIC-IV or linked electronic health record systems could enable a more holistic and temporally-informed feature set.

All analyses were based solely on the MIMIC-III dataset, which, while rich and extensively used, represents data from a single US-based academic hospital, which may restrict the generalisability of our findings to broader or international healthcare settings. The inclusion of external datasets for model testing would be crucial for evaluating performance robustness across institutional, geographic and demographic variations. Unfortunately, access to equivalent, labelled ICU datasets with comparable variables was beyond the scope of this study, but this remains a critical direction for future research. Similarly, due to the retrospective nature of the dataset, the timing and availability of variables are not controlled, and the potential for residual confounding from unmeasured or unrecorded variables remains.

We acknowledge that the analytic cohort (n=600) is modest relative to the full MIMIC-III resource and that model overfitting is a risk, especially for highly flexible ensemble methods. To mitigate overfitting, we adopted stratified threefold cross-validation, performed imputation within training folds (to prevent leakage), applied class weighting where supported and tuned model complexity (tree depth, number of estimators, regularisation parameters) using inner validation heuristics. Despite these precautions, the narrow CIs for top models are consistent with strong discriminative ability in this cohort but do not rule out optimistic performance due to cohort selection. Accordingly, we recommend external validation on independent ICU cohorts (ideally multicentre) prior to clinical deployment. Finally, although the study included several robust ML models, not all possible algorithms or hyperparameter tuning strategies were explored. Future studies could expand on this work using multicentre data and broader algorithmic comparisons.

CONCLUSION

The results showed that LR had the lowest prediction accuracy, while ETs had the highest. The ET algorithm's decision-making process was clearly shown by the SHAP and LIME explainability methodologies, emphasising the importance of specific features in mortality prediction.

ML-based algorithms can help healthcare practitioners identify patients who have a high risk of mortality and enable prompt interventions to improve patient outcomes. The interpretability and transparency offered by the explainability approaches used in this work enable clinicians to vouch for the model's judgements and have faith in its forecasts. In order to improve patient outcomes and prioritise mortality risk factors, important traits can guide future research and clinical treatments.

Future research may examine how these algorithms function when applied to larger datasets and in medical settings. Additional ML algorithms and explainability techniques may be studied in order to learn more about how to forecast death. Future study is necessary to validate the effectiveness of these algorithms on larger datasets and in healthcare settings. Clinical decision-making systems that use ML algorithms may potentially improve patient outcomes and lower healthcare costs.

This study illustrates the potential of ML methods for mortality rate prediction using the MIMIC-III dataset. The explainability approaches used in this work offer important insights into the models' decision-making process, enabling doctors to confirm the models' choices and have faith in their forecasts.

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